P-TH ROOTS IN SEPARABLE EXTENSIONS: AN ELEMENT WHOSE MINIMAL POLYNOMIAL HAS P-TH-POWER COEFFICIENTS IS A P-TH POWER

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The statement's hypothesis that L/K is separable enters only through the distinct-roots clause of (2b); the lemma feeds the p-basis machinery later in the chapter.

Editorial note. The label lemma-pth-root names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Same chapter: lemma-take-pth-root (if alpha is not a p-th power in L then T^p - alpha is irreducible over L); separability of L/K gives the minimal polynomial of alpha pairwise distinct roots

Source reference. Original reference 09HF: sources/stacks-fields/pinned-original-source.tex, source lines 1834–1834. Prerequisite pointer invoked at 3985; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let K be a field of characteristic $p > 0$. Let L/K be a separable algebraic extension. Let $\alpha \in L$.*

(1) *If the coefficients of the minimal polynomial of α over K are pth powers in K then α is a pth power in L.*
(2) *More generally, if $P \in K[T]$ is a polynomial such that (a) α is a root of P, (b) P has pairwise distinct roots in an algebraic closure, and (c) all coefficients of P are pth powers, then α is a pth power in L.*

Proof. It follows from the definitions that (2) implies (1). Assume P is as in (2). Write $P(T) = \sum_{i=0}^{d} a_i T^{d-i}$ and $a_i = b_i^p$. The polynomial $Q(T) = \sum_{i=0}^{d} b_i T^{d-i}$ has distinct roots in an algebraic closure as well, because the roots of Q are the pth roots of the roots of P. If α is not a pth power, then $T^p - \alpha$ is an irreducible polynomial over L (Lemma 09HF). Moreover Q and $T^p - \alpha$ have a root in common in an algebraic closure $\overline{L}$. Thus Q and $T^p - \alpha$ are not relatively prime, which implies $T^p - \alpha | Q$ in $L[T]$. This contradicts the fact that the roots of Q are pairwise distinct. □

Date: stacks-project:fields.tex:local-snapshot-20261012; GNU FDL 1.2 or later.